HDOC DAY – 70

Algorithm

1. Start.
2. Read a non-negative integer n.
3. Call the function digitalRoot(n).
4. Declare a static variable count and increment it.
5. If n < 10, display the recursive call count and return n.
6. Otherwise, calculate the sum of digits using % and /.
7. Recursively call digitalRoot(sum).
8. Display the digital root.
9. Stop.

C PROGRAM:

```
#include <stdio.h>


int digitalRoot(int n)
{
   static int calls = 0;
   int sum = 0;


   calls++;


   if (n < 10)
   {
      printf("Number of recursive calls: %d\n", calls);
      return n;
```

```
    }

    while (n > 0)
    {
        sum = sum + (n % 10);
        n = n / 10;
    }

    return digitalRoot(sum);
}

int main()
{
    int n, root;

    printf("Enter a number: ");
    scanf("%d", &n);

    root = digitalRoot(n);

    printf("Digital root: %d\n", root);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Input	Calculation	Digital Root	Function Calls
1	9875	9875 → 29 → 11 → 2	2	4
2	1234	1234 → 10 → 1	1	3
3	999	999 → 27 → 9	9	3
4	45	45 → 9	9	2
5	7	Already single digit	7	1